Spring 2026

LAB TASK - 05

Course Title: Operating System Lab
Course Code: CSE 2306

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Submission Date: *09-03-2026*

CODE:

```
#include<stdio.h>

void roundRobin(int p[], int bt[], int n, int tq, int ct[], int *time)
{
   int rt[20], i, done;

   for(i=0;i<n;i++)
      rt[i] = bt[i];

   while(1)
   {
      done = 1;

      for(i=0;i<n;i++)
      {
         if(rt[i] > 0)
         {
            done = 0;

            if(rt[i] <= tq)
            {
               *time += rt[i];
               ct[p[i]] = *time;
               rt[i] = 0;
            }
            else
            {
               *time += tq;
               rt[i] -= tq;
            }
         }
      }

      if(done == 1)
         break;
   }
}

void fcfs(int p[], int bt[], int n, int ct[], int *time)
{
```

```
    int i;

    for(i=0;i<n;i++)
    {
        *time += bt[i];
        ct[p[i]] = *time;
    }
}

int main()
{
    int n,i;
    int bt[20],type[20];
    int ct[20],wt[20],tat[20];

    int sysP[20],userP[20];
    int sysBT[20],userBT[20];

    int s=0,u=0;
    int tq,time=0;

    printf("Enter number of processes: ");
    scanf("%d",&n);

    printf("Enter Time Quantum: ");
    scanf("%d",&tq);

    for(i=0;i<n;i++)
    {
        printf("\nBurst Time of P%d: ",i);
        scanf("%d",&bt[i]);

        printf("Process Type (1=System,0=User): ");
        scanf("%d",&type[i]);

        if(type[i]==1)
        {
            sysP[s]=i;
            sysBT[s]=bt[i];
            s++;
        }
        else
        {
            userP[u]=i;
```

```
        userBT[u]=bt[i];
        u++;
      }
    }

    /* Queue 1 → Round Robin */
    roundRobin(sysP,sysBT,s,tq,ct,&time);

    /* Queue 2 → FCFS */
    fcfs(userP,userBT,u,ct,&time);

    float avgWT=0, avgTAT=0;

    printf("\nProcess\tBT\tWT\tTAT\n");

    for(i=0;i<n;i++)
    {
      tat[i] = ct[i];
      wt[i] = tat[i] - bt[i];

      avgWT += wt[i];
      avgTAT += tat[i];

      printf("P%d\t%d\t%d\t%d\n",i,bt[i],wt[i],tat[i]);
    }

    printf("\nAverage Waiting Time = %.2f",avgWT/n);
    printf("\nAverage Turnaround Time = %.2f\n",avgTAT/n);

    return 0;
}
```

OUTPUT:

```
Enter number of processes: 4
Enter Time Quantum: 2

Burst Time of P0: 3
Process Type (1=System,0=User): 1

Burst Time of P1: 2
Process Type (1=System,0=User): 0

Burst Time of P2: 5
Process Type (1=System,0=User): 1

Burst Time of P3: 1
Process Type (1=System,0=User): 0

Process BT      WT      TAT
P0      3       2       5
P1      2       8       10
P2      5       3       8
P3      1       10      11

Average Waiting Time = 5.75
Average Turnaround Time = 8.50

Process returned 0 (0x0)   execution time : 42.062 s
Press any key to continue.
```